

PSBs Hackathon Series 2026

(An Initiative of Government of India, Ministry of Finance , Department of Financial Services)



PRESENTS



INNOVATION | DEVELOPMENT | EXECUTION | ADVANCEMENT

Ai - C S P A R C

AI Powered Innovation for Customer: Security, Privacy, Assurance, Reliance, Convenience

Digital Trust | Ease of doing Business

Promoting Innovation and Fostering Collaboration



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

PROBLEM STATEMENT & TEAM DETAILS

TEAM NAME (AS SUBMITTED ON PORTAL)		Queen Mode	
PROBLEM STATEMENT TITLE		Unified Customer Complaint Communication Dashboard	
	NAME	GENDER	AREA OF EXPERTISE
Member 1	Shakshi Yadav	Female	AI & Backend Developer
Member 2	Sonalika Chandra	Female	AI & Frontend Developer
Member 3	Sanskriti Kulkarni	Female	AI & ML

Unified Customer Complaint Communication Dashboard

Our solution

AI-powered complaint intelligence platform that collects complaints from multiple channels (such as email, branch logs, chat, call center notes, and social media), then automatically:

- Categorizes complaints,
- Detects sentiment and urgency,
- Identifies duplicate or repeated issues,
- Generates suggested responses,
- Tracks SLA and escalation,
- Shows trends by branch, product, region, and complaint type.

How it address the problem?

- ❖ Faster resolution
- ❖ Provide live complaint status to customer.
- ❖ Identify repeated issues or duplicate complaints
- ❖ Ensure SLA compliance.
- ❖ Generating AI draft responses.

Innovation

- Root Cause Intelligence
- Complaint Spike Prediction
- Auto Action Recommendation Engine
- Customer Churn Risk Detection
- End-to-End Intelligent Complaint Lifecycle

Detailed Working Solution



Dashboard Visualization



OUTLINE OF UNIQUE & INNOVATIVE SOLUTION

“From complaint tracking to intelligent complaint resolution”

Our platform goes beyond complaint logging — it classifies, summarizes, verifies, and optimizes complaint handling using AI.

Root Cause Intelligence

AI identifies the underlying reason behind repeated complaints.

Example: 80% of ATM failure complaints linked to 3 branches.

Complaint Spike Prediction

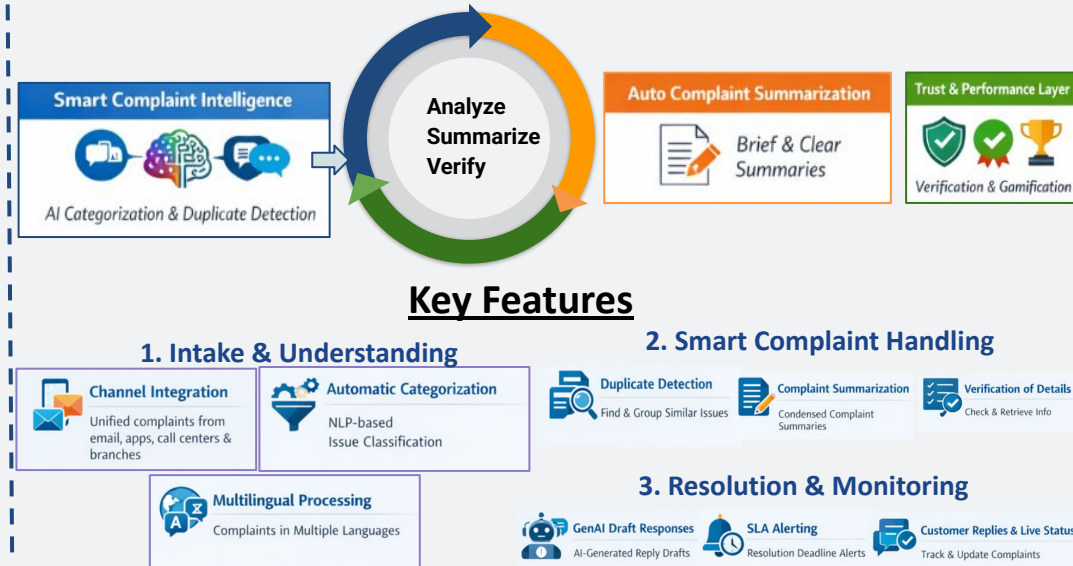
AI predicts future complaint surges before they happen.

This helps the bank act early and prevent service issues.

AI Action Recommendations

The system not only shows insights, but also suggests next actions.

Example: increase staff, fix server, notify affected customers.



70%

Faster Triage

50%

Less Duplicates

3X

Quicker Reviews

24/7

Monitoring

TECHNICAL APPROACH

Technologies Used

Frontend: React.js, Tailwind CSS

Backend: Flask / FastAPI

Database: MongoDB / PostgreSQL

AI/NLP: Python, spaCy, Transformers

GenAI: OpenAI / LLM APIs

Visualization: Chart.js / Power BI

Deployment: Docker / Cloud (AWS)

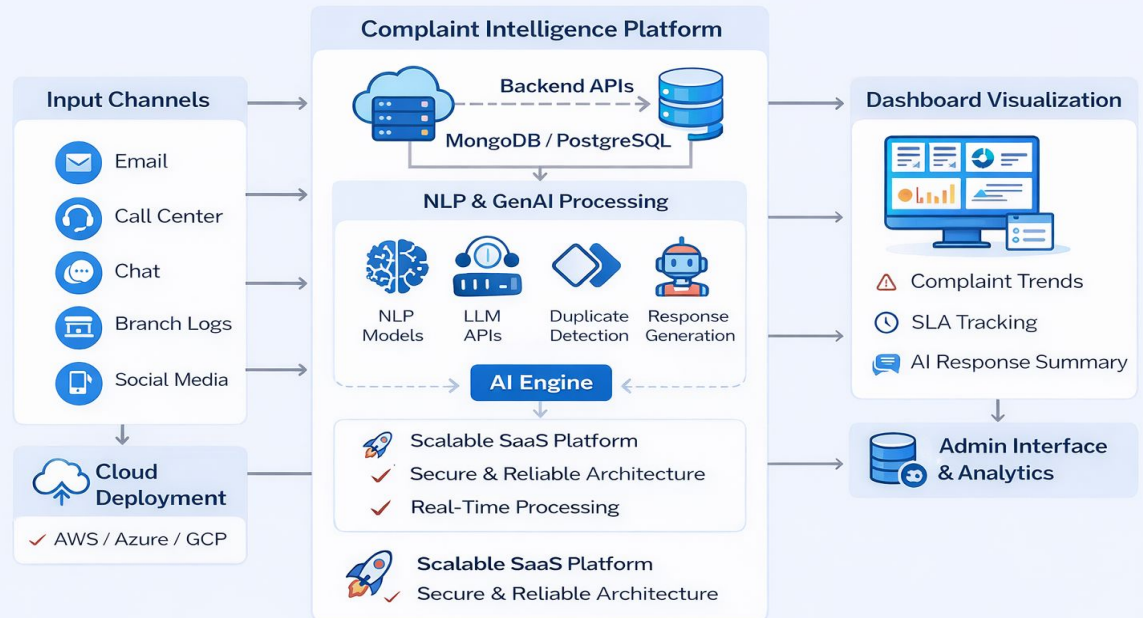
System Workflow



Prototype:

- ✓ Web dashboard
- ✓ Real-time complaint tracking
- ✓ AI-generated responses

Detailed Architecture





FEASIBILITY & VIABILITY

Technically & Operationally Feasible

- Built using **existing NLP & GenAI technologies**, reducing development complexity
- Can operate using **synthetic or anonymized complaint data**, avoiding real data dependency
- Easily deployable on **cloud infrastructure (AWS / Azure / GCP)**
- Modular architecture allows **easy integration with existing banking systems**
- Scalable design supports **high-volume real-time complaint processing**

Potential Challenges & Risks

- **Data Privacy & Security Risks**
Handling sensitive customer information
- **Multilingual complaint handling**
- **Incorrect Classification / Bias**
NLP models may misinterpret complaint context
- **Unstructured & Noisy Data**
Complaints from different channels vary in quality
- **GenAI Hallucination Risk**
AI-generated responses may be inaccurate
- **Integration Complexity**
Connecting with legacy banking systems

Mitigation Strategies

- **Data Anonymization & Encryption**
- Apply **multilingual NLP models**
- **Human-in-the-Loop Validation**
Critical cases reviewed before final response
- **Hybrid Approach (AI + Rules)**
Combines ML with rule-based logic for accuracy
- **Model Fine-Tuning & Continuous Learning**
- **API-Based Integration Layer**
Enables smooth connection with banking systems

IMPACT & BENEFITS

Before Our Solution

- Fragmented complaint handling across multiple channels
- Slow resolution
- Lack of transparency for customers
- Inefficient operations for banks
- Higher operational costs
- Lower customer satisfaction

With Our Solution

- Faster complaint resolution
- Transparent tracking for customers
- Multilingual communication support
- Reduced processing time
- Improved operational efficiency
- Better monitoring for management
- Reduced duplicate complaints
- Higher agent productivity
- Improved customer satisfaction

Benefits

01

Social

Better customer trust and reduced frustration.

02

Economic

Lower operational cost and fewer repeated service failures

03

Organizational

More efficient complaint handling and reporting

BUSINESS MODEL

Scalable AI-powered SaaS solution for intelligent complaint management

Business Model Overview

- **SaaS-based platform** for banks and financial institutions
- Subscription pricing (monthly/yearly plans)
- Tier-based features (Basic, Pro, Enterprise)
- Additional revenue from **customization & integration services.**

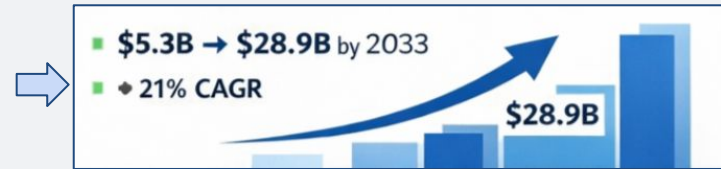
Target Customers

- Banks & NBFCs
- Fintech companies
- Insurance & financial service providers
- Large enterprises with customer support systems

Value Proposition

- Unified complaint handling
- Reduced operational costs
- Improved customer satisfaction
- AI-driven insights & automation
- Efficient SLA monitoring

Market Opportunity



Commercialization Strategy

- Faster complaint resolution
- Reduced operational costs
- Improved customer satisfaction
- AI-driven insights & automation
- Regulatory compliance support

Scalability & Expansion

- Cloud-Based Deployment
- API Integration
- Multi-Language Support
- Data Privacy & Compliance

Go-To-Market Strategy



RESEARCH & REFERENCES

1. NLP for Complaint Classification:

- <https://www.ijraset.com/research-paper/complaint-classification-model-using-nlp>
- <https://www.researchgate.net/publication/363028608>

2. AI in Customer Support Systems:

- <https://ijcttjournal.org/archives/ijctt-v71i3p101>
- https://ijsra.net/sites/default/files/fulltext_pdf/IJSRA-2023-0698.pdf

3. Complaint Management Platforms:

- <https://www.sentisum.com/customer-service-analytics-software/customer-complaint-analysis-tool>

4. Duplicate Detection & Pattern Recognition:

- <https://arxiv.org/abs/2308.11138>

5. Sentiment Analysis in Customer Feedback:

- https://link.springer.com/article/10.1007/s10044-023-01182-8?utm_source=chatgpt.com

6. Automated Complaint Handling Systems

- <https://arxiv.org/abs/2404.18963>

One Page Summary ->

https://drive.google.com/file/d/1nfmKgNjeGxp25df-LDyMCUMj_AoQ1T9w/view?usp=sharing

YouTube Video ->

<https://youtu.be/v6q-akKrHec>